ModelIt!

Student Activity Pack

Solar vs. Fossil: The Energy Showdown

Which Power Source Wins — and What's the Tradeoff?

Grade 05L2 | NGSS 5-PS3-1, 5-ESS3-1 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Where does electricity come from when you flip a light switch?

2 What do you think happens when coal is burned at a power plant?

3 Have you seen solar panels in your neighborhood? What do they do?

4 Why do some people want to use solar energy instead of burning coal or gas?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Sunlight Hours | Fossil Fuel Burn Rate | Solar Energy Output | CO2 Emissions | Air Quality Index

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Mixed Energy

Both sources at 50%

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: All Solar

Fossil Fuel at 0%, Sunlight at 80%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Cloudy Day Crisis

Sunlight at 10%, Fossil Fuel at 0%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Wind Energy — Electricity generated from wind turbines as another clean source

Energy Storage — Batteries that save solar energy for nighttime use

Electricity Demand — How much energy the community needs at any given time

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Can solar energy replace fossil fuels, or do we need both? Use evidence from your simulation to support your answers.

1. **Why can't we just switch to 100% solar energy overnight?**

2. **What would happen to air quality in cities if all cars and power plants switched to solar?**

3. **How is your town's energy mix similar to or different from the model you built?**

4. **What surprised you most about the relationship between fossil fuels and air quality?**

5. **If you were the mayor, what would your energy plan look like?**

8 STEM Challenge: Design a Clean Energy Plan for Your Town

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YOUR ENGINEERING MISSION

Design an energy plan that keeps the lights on while minimizing air pollution.

THE SCENARIO

Your town currently runs on 100% fossil fuels. The mayor has asked your team to create a plan that reduces CO_2 emissions by at least 50% without causing blackouts.

Engineering Design Thinking

1. WHAT PERCENTAGE OF ENERGY WILL COME FROM SOLAR VS. FOSSIL FUELS?
2. HOW WILL YOU HANDLE ENERGY NEEDS AT NIGHT OR DURING STORMS?
3. WHERE WILL YOU PLACE SOLAR PANELS FOR MAXIMUM SUNLIGHT?
4. HOW WILL YOU MEASURE WHETHER YOUR PLAN ACTUALLY REDUCES EMISSIONS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Renewable Energy Engineers design and install solar, wind, and other clean energy systems. They help communities transition away from fossil fuels. They earn $75,000-$130,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 In the energy model, what happens to CO2 Emissions when Fossil Fuel Burn Rate is set to 0% and Sunlight Hours are at 80%?

- **A.** CO2 Emissions increase because solar panels release CO2
- **B.** CO2 Emissions stay the same because energy is still being produced
- **C.** CO2 Emissions drop to near zero because solar energy produces no CO2
- **D.** CO2 Emissions increase because more energy is needed from backup sources

2 What is the main tradeoff between solar energy and fossil fuel energy revealed by the model?

- **A.** Solar energy is more expensive but produces more CO2
- **B.** Fossil fuels are cleaner but less reliable
- **C.** Solar energy is clean but depends on sunlight; fossil fuels are reliable but produce CO2 that harms air quality
- **D.** There is no tradeoff; solar energy is better in every way

3 What type of relationship exists between Fossil Fuel Burn Rate and Air Quality Index?

- **A.** Positive: more fossil fuels improve air quality
- **B.** No relationship: fossil fuels don't affect air
- **C.** Negative: burning more fossil fuels releases CO2 and pollutants that reduce air quality
- **D.** Positive: burning fuels cleans the air by consuming oxygen

4 During the Cloudy Day Crisis scenario (Sunlight at 10%, Fossil Fuel at 0%), what problem does the model reveal?

- **A.** CO2 Emissions spike because clouds trap greenhouse gases
- **B.** Solar Energy Output crashes and very little total energy is produced
- **C.** Air Quality drops because clouds contain pollution
- **D.** Temperature rises because clouds trap heat

5 A city switches from 100% fossil fuels to 50% solar and 50% fossil fuels. Based on the model, what would you predict?

- **A.** CO2 Emissions would stay the same because total energy hasn't changed
- **B.** Air Quality would get worse because solar panels create pollution
- **C.** CO2 Emissions would decrease by about half, and Air Quality would improve
- **D.** Energy output would double because two sources are better than one